

Student's Full Name:**Group****Text**

One of classification of Materials is solid materials can be grouped into three basic classifications: metals, ceramics and polymers. This classification is based primarily on chemical makeup and atomic as well as molecular structure. Most materials fall into one distinct grouping, although there are some intermediates. More engineering components are made of metals and alloys than of any other class of solid. But increasingly, polymers are replacing metals, because they offer a combination of properties more attractive to designers.

New ceramics are developed worldwide, which will permit materials engineers to devise more efficient heat engines and lower friction *bearings*. Ceramics have been found that become super conducting (showing electrical conductivity with very limited resistance) at extremely low temperatures (about 100 K, approximately minus 170 °C). If this phenomenon is ever achieved at *ambient temperature*, it may increase the use of ceramics and revolutionize electronics.

The best properties of materials can be combined to make composites which often combine two or more materials from these three basic classes. In high-technology applications, a new classification called advanced or smart materials emerges. These materials are semiconductors, biocompatible materials. Natural materials like wood or leather should also be mentioned, since they offer properties that, even with the innovations of today's materials scientists, are hard to beat. (From Callister and Ashby/Jones, modified and abridged)?

Task One Read the text then decide whether the statements are true or false. Correct the false statements. (5pts)

1. A material's property can be physical, chemical or mechanical
.....
2. Polymers belong to a distinct material group.
.....
3. Ceramics will increasingly be used for applications in electronics because of their hardness.
the use of ceramics and revolutionize electronics.....
4. Man-made materials are superior to natural materials.
.....

Task Two: find in the text words that correspond to the definitions given below (4pts)

- 1- Coming between two things, place or time.....
- 2- Any of the class of natural or synthetic substances composed of very large molecules.....
- 3- An inorganic solid hardened by heating to high temperature.
- 4- A solid substance that has conductivity and insulator such as germanium and silicon.....

Task Three: circle the correct quantifier to complete the sentence (5pts)

- 1- How (**many** / **much**) Valence Electrons are in the outer energy level of Phosphorus?
- 2- The force of attraction between particles of (**few** / **less**) substances in the increasing order is Oxygen > water < Sugar.
- 3- The job is great. Even if I have (**much** / **many**) work to do, I have (**less** / **fewer**) stress and (**less** / **fewer**) problems.
- 4- (**much** / **many**) carbon dioxide is increased in the atmosphere because of human activities.

Task Four: Fill in the blanks with the words in the box (5pts)

Physical	Narrowly	Daily	components	substance
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Materials is synonymous withthat come across onbasis.
Sometimes, it is used more to refer towith certain
.....properties which are used as inputs to production or manufacturing.

Good Luck

2,2.

Key Answer

Task One Read the text then decide whether the statements are true or false. Correct the false statements. (5pts)

5. A material's property can be physical, chemical or mechanical (**True**)
6. Polymers belong to a distinct material group. (**True**)
7. Ceramics will increasingly be used for applications in electronics because of their hardness. (**False**) because it is super conductingelectrons)
8. Man-made materials are superior to natural materials. (**False**) Natural materials like wood or leather should also be mentioned, since they offer properties that, even with the innovations of today's materials scientists, are hard to beat.

Task Two: find in the text words that correspond to the definitions given below (4pts)

- 5- Coming between two things, place or time.....intermediate...
- 6- Any of the class of natural or synthetic substances composed of very large molecules Polymer
- 7- An inorganic solid hardened by heating to high temperature. ...ceramic.....
- 8- A solid substance that has conductivity and insulator such as germanium and silicon semi-conductor.....

Task Three: circle the correct quantifier to complete the sentence (5pts)

- 5- How (**many**) Valence Electrons are in the outer energy level of Phosphorus?
- 6- The force of attraction between particles of (**few**) substances in the increasing order is Oxygen > water < Sugar.
- 7- The job is great. Even if I have (**much**) work to do, I have (**less**) stress and (**fewer**) complications.
- 8- (**much**) carbon dioxide is increased in the atmosphere because of human activities.

Task Four: Fill in the blanks with the words in the box (5pts)

Materials is synonymous with ...substance...that come across on ...daily basis. Sometimes, it is used more narrowly.... to refer to ...components...with certain ...physical.....properties which are used as inputs to production or manufacturing.